ACHELION CAPITAL

PM OPERATIONAL PLAYBOOK

Decision Framework, Daily Protocol & Regime Management Guide

v4.0 · March 2026 · Architecture AB · 58/20/14/8 · Seven-Pillar + Seven-Gap Upgrades

v4.0 UPGRADE SUMMARY — WHAT IS NEW

v4.0 closes seven world-class institutional risk gaps: (1) Factor Exposure Monitor — detects hidden concentration in AI capex, Taiwan manufacturing, BTC beta, and dollar sensitivity before it becomes a drawdown. (2) Volatility-Adjusted Re-entry Sizing — re-entry tranche size scaled to current vol conditions so you come back smaller when vol is elevated. (3) Portfolio Drawdown Sentinel — independent of ARAS, monitors NAV vs high-water mark with automatic ceiling triggers at −8%, −12%, and −18%. (4) Liquidity-Adjusted Execution — VWAP in elevated vol, reordered priority in crisis (illiquid positions first). (5) Permanent Tail Risk Hedge — QQQ puts, always on, rolled at 30 DTE, sized for 10–15% protection in a −30% QQQ scenario. (6) Conviction Decay Function — automatic position weight reduction when a thesis underperforms for 45+ days. (7) Stress Scenario Library — daily computation of 8 defined stress scenarios against current portfolio.

1. AUTHORITY HIERARCHY — V4.0

The authority hierarchy is unchanged. Two new ceiling authorities exist but both can only make the portfolio MORE defensive, never less.

AUTHORITY	CEILING RULE	OVERRIDE?
ARAS	Absolute regime ceiling. Always prevails unless PDS is lower.	**No human override**
PDS — NEW v4.0	Drawdown ceiling — fires when portfolio falls 12% or 18% from HWM. Lower of ARAS or PDS always prevails.	PM can override with documented justification
FEM — Advisory	Factor concentration monitor. No ceiling authority. Advisory signal to PM.	N/A — advisory only
PTRH — Additive	QQQ puts — outside exposure ceiling calculation. Never removed below DEFENSIVE.	PM can suspend with GP approval
PM + GPs	PM may override PDS ceiling with documented justification and GP co-sign. Cannot override ARAS.	PDS only

2. DAILY PM PROTOCOL — V4.0 ASSESSMENT SEQUENCE

The daily assessment sequence adds five new checks from v4.0. Complete the existing protocol first, then run the five new v4.0 checks before finalizing the session log.

STEP	CHECK	ACTION IF FLAGGED	SOURCE
1	ARAS Composite Score	Regime determination → ceiling setting	aras.py output
2	Circuit Breakers	Hard trigger → immediate deleverage per kill chain	circuit_breaker.py
3	SLOF Overlay Status	Confirm SLOF on/off per regime	slof.py output
4	Escalation Rule Status	Check auto-suppress condition	escalation_engine.py
5 ★	Factor Exposure Monitor — NEW v4.0	WATCH: document. ALERT: mandatory review within 24h	factor_exposure.py

STEP	CHECK	ACTION IF FLAGGED	SOURCE
6 ★	Portfolio Drawdown Sentinel — NEW v4.0	WARNING: heightened monitoring. DELEVERAGE: automatic execution	drawdown_sentinel.py
7 ★	Tail Hedge Status — NEW v4.0	DTE ≤ 30: initiate roll. Regime shift: resize per PTRH schedule	tail_hedge.py
8 ★	Stress Scenario P&L — NEW v4.0	Advisory. Worst scenario noted in session log.	stress_scenarios.py
9 ★	Conviction Decay Flags — NEW v4.0	Day 45: acknowledge. Day 90: sizing review. Day 135: full review	perm.py CDF
10	Session Log + LP Narrative	Document all decisions, overrides, and flags	pm_protocol.py

3. FACTOR EXPOSURE MONITOR — PM DECISION GUIDE

The FEM tells you what your regime model cannot: whether you are running what looks like a diversified portfolio but is effectively a single concentrated bet. Review the FEM output every morning as part of Step 5.

3.1 When FEM Shows WATCH (>65% single factor)

Document in the session log. No mandatory action is required. Consider whether the concentration is intentional (it is your thesis) or accidental (positions have correlated more than expected). If intentional and within conviction parameters, note it and move on. If accidental, flag for the next portfolio review.

3.2 When FEM Shows ALERT (>80% single factor)

Mandatory review within 24 hours. Your choices are: (A) Confirm the concentration is deliberate and within your risk tolerance — document a conviction reaffirmation. (B) Identify a trim that reduces the factor exposure below the ALERT threshold. (C) Accept the advisory +0.05 composite input to ARAS, which may shift the regime toward WATCH if you are near a threshold.

WHAT HIGH FACTOR EXPOSURE ACTUALLY MEANS

A portfolio showing 82% AI Capex factor exposure is not necessarily wrong — if your thesis is AI infrastructure, concentration is the point. The FEM ALERT is not telling you to diversify. It is telling you that in a scenario where the AI capex cycle pauses — a single earnings miss from a major hyperscaler, a Taiwan geopolitical event, a regulatory overhang — your portfolio has minimal natural offset. That risk is known and accepted. ALERT simply means it must be acknowledged explicitly, not ignored.

4. PORTFOLIO DRAWDOWN SENTINEL — PM DECISION GUIDE

The PDS is the only new v4.0 module with ceiling authority. It operates independently of ARAS. It protects investors when ARMS itself is wrong or slow — and any honest system must plan for that possibility.

4.1 WARNING Level (−8% from HWM)

No mandatory action. Heightened monitoring. Review all open positions. Confirm conviction levels are current. Check whether any Conviction Decay Function flags are active. Consider whether market conditions have changed in ways that make your thesis weaker. Document the review in the session log.

4.2 DELEVERAGE 1 (−12% from HWM)

The system automatically enforces a 60% gross ceiling. This happens regardless of ARAS regime. If ARAS says RISK_ON at 100%, and PDS fires at −12%, the effective ceiling is 60%. The kill chain executes to 60% using the standard priority order with liquidity-adjusted execution. The PM is notified immediately. This is not an override request — it is an automatic execution.

4.3 DELEVERAGE 2 (−18% from HWM)

The system automatically enforces a 30% gross ceiling. LP communication is triggered — the auto-narrative engine generates a disclosure. The PM must review and approve the narrative before it is sent, but the deleverage itself is automatic. GP notification is required within 2 hours. This level is designed to never be reached if the system is operating correctly — it is the final backstop.

PDS LEVEL	AUTOMATIC	PM ACTION REQUIRED	GP NOTIFICATION
WARNING	None — advisory	Review positions. Document.	Not required
DELEVERAGE 1	60% ceiling enforced. Kill chain executes.	Review and confirm execution. Update session log.	Notify within 24h
DELEVERAGE 2	30% ceiling enforced. LP narrative triggered.	Approve LP narrative before send. Full position review.	**Notify within 2h**

5. PERMANENT TAIL RISK HEDGE — PM OPERATIONAL GUIDE

5.1 The Two Layers — Permanent and Dynamic

There are two distinct hedge layers that coexist at all times. The permanent PTRH is always on — it never goes to zero and does not depend on regime. The dynamic STRC reserve layer deploys into the same instrument (QQQ puts) but is separately managed per the existing STRC protocols. Both layers are additive. In DEFENSIVE regime, you have both layers fully deployed simultaneously.

LAYER	ALWAYS ON?	SIZED BY	REGIME DEPENDENCY
Permanent PTRH	**YES — never removed**	Target payoff: 10–15% protection in −30% QQQ	Sizes up in WATCH/DEFENSIVE. In CRASH: hold, no new buys.
Dynamic STRC Reserve	**No — deploys NEUTRAL+**	STRC sleeve size per existing protocol	Locks in RISK_ON. Deploys 50% NEUTRAL, 100% DEFENSIVE/CRASH.

5.2 Roll Protocol — 30 DTE Trigger

When QQQ put positions reach 30 days to expiry (DTE), initiate a roll. The roll involves buying the next expiry at the same strike percentage (10–15% OTM relative to current QQQ price) and selling the expiring position. Target the 60–90 day expiry. If implied volatility is elevated at roll time (VIX >35), consider rolling to a longer expiry (90 days) to reduce the vol premium paid.

5.3 Why QQQ Puts, Not SPX

This is documented here for LP conversations. QQQ puts hedge the technology sector directly. In the stress scenarios most relevant to Achelion's portfolio — AI capex pause, Taiwan manufacturing disruption, tech sector rotation — QQQ falls materially harder than SPX. In 2022, QQQ fell 32.6% while SPX fell 18.1%. An SPX put would have provided roughly 55% of the protection of a QQQ put per dollar of premium in that scenario. The portfolio's AI concentration means it will behave more like QQQ than SPX in both directions.

6. VOLATILITY-ADJUSTED RE-ENTRY — PM DECISION GUIDE

When ARES clears all four gates and PM approval is given, VARES determines the size of each re-entry tranche based on current volatility. The PM does not set the tranche size — VARES calculates it automatically. The PM confirms the calculation and approves execution.

THE KEY PRINCIPLE

Re-entry size is inversely proportional to current volatility. When VIX is elevated relative to its 90-day average, you come back smaller. This is not timidity — it is correct position sizing. A portfolio re-entering at 57% of normal tranche size when VIX is at 35 is protecting itself against a second leg down that has not yet

arrived. If the regime holds and vol normalizes, the subsequent tranches restore full exposure. If vol spikes again, you were right to come back small.

VIX CONDITION	VOL ADJ FACTOR	TRANCHE SIZE	PRACTICAL EXAMPLE
VIX at avg (VIX=20, avg=20)	**1.00 (full)**	33% of target	Target 60% ceiling → T1 adds 20pp to current 40%
VIX elevated (VIX=35, avg=20)	**0.57 (reduced)**	19% of target	Target 60% → T1 adds only 11pp. Cautious re-entry.
VIX high (VIX=50, avg=20)	**0.40 → floor 0.15**	15% of target (min)	Minimum tranche floor applies. Very cautious.

7. CONVICTION DECAY FUNCTION — PM DECISION GUIDE

The most dangerous behavior in concentrated portfolios is maintaining high conviction on a position that is quietly failing while the PM refuses to update their priors. The CDF makes this failure visible and automatically reduces position weight — without requiring the PM to admit they were wrong.

7.1 When a Position Enters Decay

A position enters decay when it has underperformed QQQ by more than 10pp over 45 consecutive days. This means it is not just lagging — it is significantly underperforming the technology benchmark that Achelion's thesis is built on. The first decay flag (Day 45) is a warning. It does not require action, but it requires acknowledgment in the session log.

7.2 PM Response Options

SCENARIO	RECOMMENDED ACTION	DOCUMENTATION REQUIRED
Still believe in the thesis — temporary underperformance	Document a conviction reaffirmation. Provide updated thesis rationale. Decay clock resets to zero.	Written thesis update in session log.
Thesis partially intact but weakening	Allow decay to reduce sizing naturally. Use smaller position as the thesis resolves. Do not force a sell.	Acknowledge in session log. Outline what would restore conviction.
Thesis is broken	Initiate an orderly exit via PERM protocols. Do not wait for Day 135. The CDF is a signal — PERM is the exit mechanism.	Full fundamental review documented. Exit rationale written.
Day 135 reached without resolution	Mandatory full fundamental review required. GP must be briefed. The position is sized at 60% of original C^2 weight until resolved.	**Formal review memo. GP sign-off on path forward.**

8. STRESS SCENARIO LIBRARY — PM INTERPRETATION GUIDE

The SSL output appears in the daily monitor every morning. It is advisory — it does not trigger any automatic action. Its purpose is to make the PM aware of tail exposure before a tail event occurs, not after.

8.1 How to Read the Scenario Output

Each scenario shows gross portfolio P&L (before the PTRH payoff) and net P&L (after estimated PTRH payoff). Focus on the net figure — it represents what investors would actually experience after the permanent hedge pays off. A well-structured portfolio should show net losses of less than 20% even in the worst defined scenario.

8.2 Scenarios Requiring PM Attention

IF THIS SCENARIO SHOWS...	IT MEANS...	CONSIDER...
S8 (AI Bubble −40%) shows net loss >25%	Equity sleeve concentration in AI names is too high relative to hedge size	Increasing PTRH size or trimming highest-beta AI names

IF THIS SCENARIO SHOWS...	IT MEANS...	CONSIDER...
S6 (BTC −60%) shows net loss >20%	Crypto sleeve is outsized relative to defensive offset	Whether 20% crypto allocation is appropriate for current regime
S5 (Hormuz 90d) shows net gain	Defense sleeve (SGOL, DBMF) is correctly positioned for the current geopolitical risk	No action — this is the architecture working
Worst scenario is S7 (Fed +100bps)	Rate sensitivity of the portfolio is elevated — SGOV allocation is working but may not be sufficient	Whether DBMF allocation adequately offsets rate sensitivity

9. COMPLETE REGIME REFERENCE — V4.0

REGIME	CEILING	SLOF	FEM	PTRH	VARES	CDF	PDS OVERRIDE?
RISK_ON	**100%**	**Active**	Monitor	1.0× base	N/A	Active	PDS can add ceiling independently
WATCH	**100%**	**Active**	Monitor	**1.25×**	N/A	Active	PDS can add ceiling independently
NEUTRAL	**75%**	**Reduce**	Alert→review	**1.25×+STRC**	N/A	Active	PDS can add ceiling independently
DEFENSIVE	**40%**	**Remove**	Alert→review	**1.5×+STRC**	**Engaged**	Active	PDS ceiling: 40% or lower
CRASH	**15%**	**Remove**	Alert→review	**Hold—no buys**	VARES active	Active	PDS ceiling: 15% or lower

10. LICENSING READINESS — ACHELION TECH

v4.0 closes the final institutional gap set that separates a sophisticated risk management system from a world-class one. The following table documents the licensing differentiators against competitive alternatives.

CAPABILITY	ARMS v4.0	TYPICAL $50M FUND	INSTITUTIONAL SYSTEM
Regime detection with confidence scoring	**✓ 7-module ARAS**	Partial — rules-based	✓ Bloomberg MARS
Factor exposure monitoring	**✓ FEM — v4.0**	Rarely	✓ Barra/MSCI
Portfolio drawdown sentinel (independent)	**✓ PDS — v4.0**	Manual HWM tracking	✓ Custom
Permanent tail risk hedge (structural)	**✓ PTRH QQQ puts — v4.0**	Rarely — cost concern	✓ Universa-style
Volatility-adjusted re-entry sizing	**✓ VARES — v4.0**	No	Partial
Conviction decay with automatic weighting	**✓ CDF — v4.0**	No	No
Daily stress scenario library (8 scenarios)	**✓ SSL — v4.0**	No	✓ Custom
Liquidity-adjusted crisis execution	**✓ LAEP — v4.0**	No	Partial
Automated LP narrative generation	**✓ v3.1 (unchanged)**	No	No
Fund-agnostic config layer (licensable)	**✓ v3.1 (unchanged)**	N/A — custom per fund	✓ Bloomberg

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